

Advantage Online Shopping (AOS)

Performance Test Summary Results Report

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Executive Summary

Performance testing was conducted against the Advantage Online Shopping (AOS) website in the pre-production environment using OpenText LoadRunner Professional. Testing was performed against the three core business processes identified in the approved Performance Test Plan: Account Registration, Item Selection, and Order Placement.

A baseline workload of 5 concurrent virtual users was executed (Registration:1, Item Selection:2, Order Placement:2). A total of 470 transactions were executed, with 468 successful and 2 failed transactions, resulting in a 99.57% success rate. The application remained available throughout testing, establishing a baseline benchmark for future testing.

Test Objectives Assessment

Assess performance	Achieved
Measure response times	Achieved
Measure throughput	Achieved
Measure error rate	Achieved
Measure transaction success rate	Achieved
Identify bottlenecks	Achieved
Establish baseline benchmark	Achieved

Scope Coverage: 100% (Registration, Item Selection, Order Placement all tested)

KPI Results

Average Response Time	9.008 sec
Maximum Concurrent Users	5
Average Throughput	5,026 bytes/sec
Error Rate	0.43%
Transaction Success Rate	99.57%

Business Process Results

Account Registration:

Create New Account	1.677 sec
Register User	1.498 sec
Assessment	PASS. Stable, no failures, response times below 2 seconds.

Item Selection:

Landing Page	13.727 sec
Preferred Category	14.609 sec
Select Preferred Item	18.514 sec
Add To Cart	14.223 sec
Assessment	Performance Concern Identified. Slowest business process with bottlenecks in page loading, category navigation, product selection and cart operations.

Order Placement:

Checkout	9.014 sec
Pay Now	6.536 sec
Assessment	PASS WITH OBSERVATIONS. Stable but checkout/payment should be optimized; one payment failure occurred.

HTTP Response Analysis:

HTTP200	898
HTTP201	26
HTTP304	2
HTTP404	3
HTTP409	1

Most requests succeeded. Minor 404 and 409 responses had minimal impact.

Workload Characteristics:

Max Running Vusers	5
Total Hits	930
Average Hits/sec	1.558
Passed Transactions/sec	0.784
Passed Transactions/minute	47.04

Primary Bottlenecks:

Select Preferred Item	18.514 sec
Preferred Category	14.609 sec
Add To Cart	14.223 sec
Landing Page	13.727 sec
Checkout	9.014 sec

Recommendations

1. Optimize database queries.
2. Improve API response times.
3. Implement caching.
4. Investigate HTTP404/409 responses.
5. Optimize images, CSS and JavaScript.
6. Execute higher-volume Load Testing.
7. Execute Stress Testing.
8. Execute Soak Testing.
9. Capture infrastructure metrics.

Overall Conclusion

The application successfully completed baseline testing and demonstrated excellent stability with a 99.57% success rate. Account Registration performed well, Order Placement remained stable with moderate delays, while Item Selection is the primary performance bottleneck. Performance optimization is recommended before progressing to larger-scale load, stress and endurance testing.